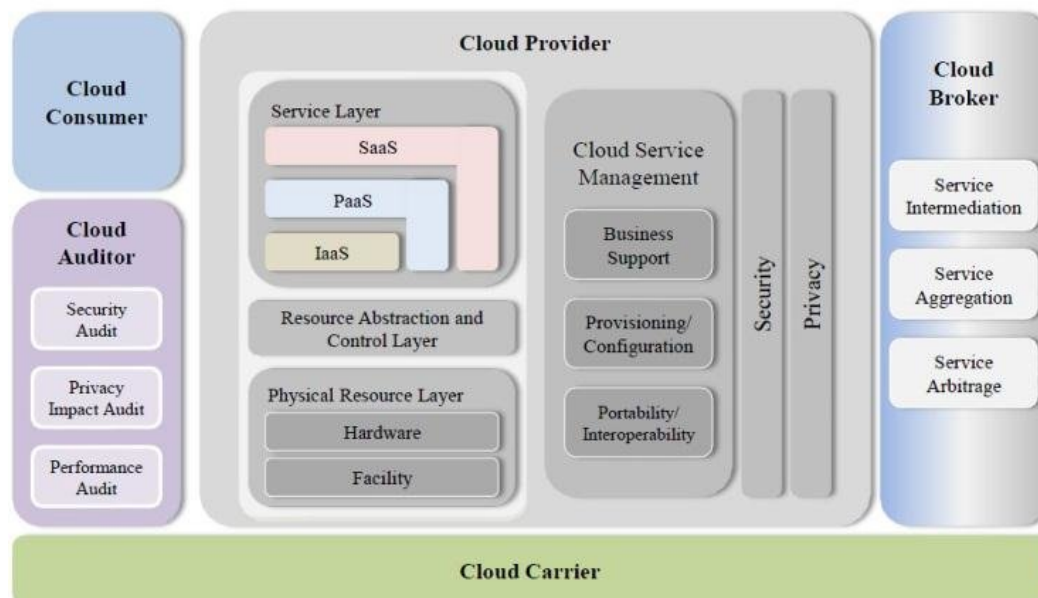


Cloud Reference Model (NIST Reference Architecture)

defines an **abstract framework** for understanding **cloud computing roles** and their **interactions**.

The model defines roles:

1. **Cloud Consumer** – Uses cloud services.
2. **Cloud Provider** – Offers cloud services.
3. **Cloud Broker** – Manages use and delivery of cloud services.
4. **Cloud Auditor** – Evaluates services for security, performance, compliance.
5. **Cloud Carrier** – Provides connectivity between consumers and providers.



The **NIST Cloud Computing Architecture** is a **standard reference model** developed by the **National Institute of Standards and Technology (NIST)** in the U.S. to define key roles and components in a cloud computing environment.

NIST SP 500-292 document,

Core Components of NIST Cloud Computing Architecture

1. Cloud Consumer

- This is the **end-user**(person) or **organization** that uses cloud services provided by the Cloud Provider.
- Example: A business using Google Workspace or AWS.
- Examples include businesses using cloud storage or individuals using cloud-based applications.

2. Cloud Provider

- The entity that offers cloud services (IaaS, PaaS, SaaS).
- Responsible for infrastructure, service provisioning, billing, security, etc.
- Example: AWS, Azure, Google Cloud.

3. Cloud Broker

- Acts as an **intermediary** between the consumer and multiple providers.
- Helps in **selection**, **negotiation**, **aggregation**, and **arbitrage** of services.
- Example: A broker can find the cheapest cloud storage option from multiple providers.

4. Cloud Auditor

- Performs **independent assessments** of cloud services for:
 - Security
 - Privacy
 - Performance
 - Compliance
- Example: Ensures GDPR compliance in a cloud deployment.
- Auditors assess whether the Cloud Provider meets regulatory requirements, security standards, and service-level agreements (SLAs). This role ensures trust and accountability in the cloud ecosystem.

5. Cloud Carrier

- Provides the **connectivity and network** that links the cloud consumer with the provider.
- It is like an Internet Service Provider (ISP).
- Example: Airtel, AT&T, Jio, etc.

Relationships and Workflow

The NIST model emphasizes how these roles interact:

- **Consumer-Provider:** The core relationship where consumers access services directly or via a Broker.
- **Broker's Role:** Enhances flexibility by managing multi-cloud environments or adding features like enhanced security or billing management.
- **Auditor's Oversight:** Ensures Providers meet compliance and performance standards, protecting Consumers.
- **Carrier's Connectivity:** Acts as the backbone, enabling seamless data flow across the ecosystem.